

GED Math Assignment 8 Solutions ①

Part 1

A. 1. $V = \frac{1}{3} Bh$

$$= \frac{1}{3}(3 \times 3)(3)$$
$$= 9 \text{ in}^3$$

3. $V = \frac{1}{3} \pi r^2 h$

$$= \frac{1}{3} \pi (5)^2 (12)$$

$$= (4)(25) \pi$$
$$= 100 \pi \text{ cm}^3$$
$$= 314 \text{ cm}^3$$

2. $V = \frac{1}{3} \pi r^2 h$

$$= \frac{1}{3} \pi (4.5)^2 (6)$$

$$= 2 \pi (20.25)$$

$$= (40.5) \pi$$

$$= 127.2 \text{ in}^2$$

4. $V = \frac{1}{3} Bh$

$$= \frac{1}{3} (12 \times 12)(10)$$

$$= 480 \text{ m}^3$$

5. $V = \frac{1}{3} Bh$

$$= \frac{1}{3} (2 \times 2)(8)$$

$$= \frac{32}{3}$$

$$= 10.7 \text{ in}^2$$

6. $V = \frac{1}{3} \pi r^2 h$

$$= \frac{1}{3} \pi (7.5)^2 (18)$$

$$= 6 \pi (56.25)$$

$$= 1060.3 \text{ cm}^3$$

7. Old Volume = $\frac{1}{3} Bh$
 $= \frac{1}{3} (10 \times 10)(15)$
 $= (5)(100)$
 $= 500 \text{ cm}^3$

New Volume = $\frac{1}{3} (14 \times 14)(15)$
 $= 5(196)$
 $= 980 \text{ cm}^3$

so New - Old = $980 - 500$
 $= 480 \text{ cm}^3$

∴ The new package will hold
 480 cm^3 more.

8. $V = \frac{1}{3} \pi r^2 h$
 $= \frac{1}{3} \pi (4)^2 (4)$
 $= \frac{64\pi}{3}$
 $\approx 21.3 \pi$
 $= 67.02 \text{ in}^3$

Part 2

A. $SA = 2(3 \times 4) + 2(3 \times 6) + 2(6 \times 4)$
 $= 24 + 36 + 48$
 $= 108 \text{ units}^2$

p2

$$A. 2. SA = \frac{1}{2} p s + B$$

$$= \frac{1}{2} (12)(5) + (3 \times 3)$$

$$= 30 + 9$$

$$= 39 \text{ in}^2$$

$$3. SA = (2 \times 2)(6)$$

$$= 24 \text{ units}^2$$

$$4. SA = \frac{1}{2} p s + B$$

$$= \frac{1}{2} (48)(8) + (12 \times 12)$$

$$= 336 \text{ m}^2$$

$$5. SA = 2(5 \times 4) + 2(4 \times 12) + 2(12 \times 5)$$

$$= 40 + 96 + 120$$

$$= 256 \text{ in}^2$$

$$6. SA = 6 \times (8 \times 8)$$

$$= 384 \text{ in}^2$$

7. $SA = 150 \text{ in}^2$

$$SA = 6 \times (L^2)$$

$$6L^2 = 150$$

$$L^2 = \frac{150}{6}$$

$$L^2 = 25$$

$$L = 5 \text{ in} \quad (A)$$

9. Old $SA = 6 \times (3 \times 3)$
 $= 54 \text{ cm}^2$

$$\text{Now } SA = 6 \times (6 \times 6)$$
$$= 216 \text{ cm}^2$$

so $\text{New} - \text{Old} = 216 - 54$
 $= 162 \text{ cm}^2$

∴ The new box's SA is 162 cm^2 greater

A2 #1. $SA = 2\pi r^2 + 2\pi rh$
 $= 2\pi(7)^2 + 2\pi(7)(5)$

$$= 98\pi + 70\pi$$
$$= 168\pi \text{ units}^2$$

$$2. SA = \pi rs + \pi r^2$$

$$= \pi(4)(6) + \pi(4)^2$$

$$= 24\pi + 16\pi$$

$$= 40\pi \text{ units}^2$$

$$3. SA = 4\pi r^2$$

$$= 4\pi(3)^2$$

$$= 36\pi \text{ units}^2$$

$$4. SA = 2\pi r^2 + 2\pi rh$$

$$= 2\pi(9)^2 + 2\pi(9)(12)$$

$$= 162\pi + 216\pi$$

$$= 378\pi \text{ units}^2$$

$$5. SA = 4\pi r^2$$

$$= 4\pi(10)^2$$

$$= 400\pi \text{ units}^2$$

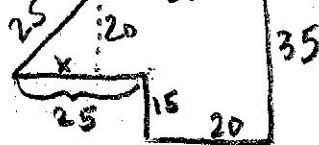
$$6. SA = \pi rs + \pi r^2$$

$$= \pi(5)(8) + \pi(5)^2$$

$$= 40\pi + 25\pi$$

$$= 65\pi \text{ units}^2$$

Part 3



$$\begin{aligned} A1. \quad P &= 30 + 35 + 20 + 15 + 25 + 25 \\ &= 150 \text{ units} \end{aligned}$$

$$\begin{aligned} x^2 + 20^2 &= 25^2 \\ x &= \sqrt{25^2 - 20^2} \\ x &= 15 \end{aligned}$$

$$\begin{aligned} 2. \quad p &= L \times W + \frac{\pi d}{2} \\ &= 10 \times 12 + \frac{\pi(10)}{2} \\ &= 120 + 5\pi \\ &= 135.7 \text{ units} \end{aligned}$$

$$\begin{aligned} A &= \frac{1}{2}(15)(20) + (10)(12) \\ &\quad + (20)(35) \\ &= 1050 \text{ units}^2 \end{aligned}$$

$$\begin{aligned} A &= \pi r^2 + L \times W \\ &= \pi(5)^2 + (12 \times 10) \\ &= 198.5 \text{ units}^2 \end{aligned}$$

$$\begin{aligned} 3. \quad P &= 12 + 6 + 17 + 10 + 9 + 3 + 15 \\ &= 72 \text{ units} \end{aligned}$$

$$\begin{aligned} A &= \frac{1}{2}(12)(3+12) + \frac{1}{2}(6)(17+9) \\ &= 90 + 78 \\ &= 168 \text{ units}^2 \end{aligned}$$

$$\begin{aligned} B. \quad 4. \quad V &= 4 \times 3 \times 5 + (2+3) \times 5 \times 5 \\ &= 60 + 125 \\ &= 185 \text{ units}^3 \end{aligned}$$

$$5. \quad V = 7 \times 7 \times 3 + \frac{1}{3}(7 \times 7)(8)$$

$$= 147 + \frac{392}{3}$$

$$= 277.67 \text{ units}^3$$

$$6. \quad V = 11 \times 11 \times 2 + \pi(5)^2 \times 2$$

$$= 242 + 50\pi$$

$$= 399.1 \text{ units}^3$$